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Dynamic and Functional miRNA-Mediated Gene Regulations in Response to Recurrent Environmental Challenges During Biological Invasions

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ABSTRACT

Biological invasions offer a valuable 'natural experiment' to investigate survival mechanisms, as invaders successfully endure substantial environmental changes during their geographical spread and settlement. Phenotypic plasticity enhances fitness by enabling rapid responses without requiring new genetic variations. Among numerous mechanisms involved in phenotypic plasticity, microRNAs (miRNAs) and their regulatory networks are believed to enable rapid responses by fine-tuning gene expression, though their roles remain poorly understood. By integrating miRNAomic and transcriptomic analyses in the model invasive ascidian *Ciona robusta*, we simulated recurring salinity stresses encountered during invasions to investigate the molecular mechanisms of miRNA-mediated gene regulation in response to recurrent environmental challenges. Multiple analyses demonstrated that miRNAs exhibited rapid, dynamic and reversible responses to recurrent stresses, displaying duration-dependent and stage-specific profiles. The upregulation of genes in the miRNA biogenesis process, rather than the decay pathway, primarily accounted for the increased expression abundance of miRNAs. Responsive miRNAs regulated target genes through an intricate regulatory network, demonstrated by both up and downregulatory relationships and diverse binding sites. Interestingly, miRNAs and their target genes exhibited a 'stress memory' effect, where miRNAs 'remembered' previous challenges and further mediated the enhanced response of target genes at later stresses. Functionally, miRNA-mediated salinity coping strategies and associated genes exhibited temporal variations depending on challenge duration and stage, and these strategies primarily involved the modulation and alternation of free amino acid metabolism and ion transport to maintain osmotic homeostasis. These findings highlight the importance of miRNA-mediated regulatory networks in shaping short-term phenotypic plasticity in response to environmental changes.

1 | Introduction

Biological invasions are widely recognised as a major environmental and ecological concern across various ecosystems, causing substantial economic and ecological losses globally (Briski et al. 2018). The large-scale negative impacts underscore the critical need to understand the mechanisms driving invasion success. Throughout the whole invasion process, invaders frequently encounter recurrent harsh environmental challenges that jeopardise their survival and ultimately determine the outcomes of biological invasions (Li et al. 2020; Huang, Li, Shenkar, and Zhan, 2023). Rapid response to recurrent environmental stimuli is crucial for the success of invasions (Ni et al. 2018; Huang, Li, Shenkar, and Zhan, 2023). This is especially important when species must navigate multiple barriers, such as the cultivation at the introduction stage and the survival at the establishment stage (Zhan et al. 2015; Li et al. 2020). Therefore, a comprehensive investigation into the mechanisms behind rapid responses to environmental challenges is central to understanding invasion success.

Phenotypic plasticity, which refers to the change in phenotypic expression of a genotype in response to environmental factors, has been shown to contribute to invasion success by increasing the fitness of invaders in harsh environments (Richards et al. 2006). From a molecular perspective, the mechanisms leading to increased fitness are highly complex, involving multiple processes such as gene expression differentiation, epigenetic regulation and alternative splicing flexibility, as well as their dynamic interactions (Huang, Li, Shenkar, and Zhan, 2023). Among these, epigenetic processes are crucial components, particularly for rapid responses to environmental challenges (Bossdorf et al. 2008). Without altering the DNA nucleotide sequences, epigenetic processes can react faster than genetic variations to mitigate the adverse effects of environmental changes on invaders (Jonsson and Jonsson 2019). Several epigenetic processes, including DNA methylation, histone modification and non-coding RNA regulation, have been reported as drivers contributing to phenotypic plasticity (Kooke et al. 2015). Among non-coding RNAs, microRNAs (miRNAs) can interact with both coding (e.g., messenger RNAs, mRNAs) and other non-coding RNAs (e.g., long non-coding RNAs, lncRNAs) to regulate gene expression, thereby influencing various phenotypic and behavioural traits for rapid environmental response (Paraskevopoulou and Hatzigeorgiou 2016; Sunkar et al. 2012).

It is well-known that miRNAs are a vital component of gene expression regulatory networks, post-transcriptionally tuning the expression of more than one-third of protein-coding genes (Friedman et al. 2009). Typically, miRNAs load into the Argonaute (AGO) complex and then specifically bind to the 3′ untranslated regions (3′UTR) of target mRNAs, leading to the repression of target genes through a combination of mRNA degradation and translation repression (Bartel 2009). Interestingly, emerging evidence shows that miRNA binding domains can also include not only the 3′UTR but also many other diverse regions, such as 5′UTR, coding region, promoter and enhancer of target genes (Pu et al. 2019; Xiao et al. 2017). Studies have illustrated that multiple miRNA binding domains can diversify post-transcriptional regulations of gene expression, such as fine-tuning the timing and magnitude of post-transcriptional regulation, regulating protein abundance based on or independent of the cellular state and controlling protein abundance in alternative splice variants (Pu et al. 2019; Wu et al. 2013). Apart from their well-known role in repressing gene expression, miRNAs can even promote the expression of target genes (Pu et al. 2019; Xiao et al. 2017). For instance, some miRNAs can guide the AGO complex to bind with AU-rich elements, thereby promoting the translation of their target genes (Vasudevan 2012). Similar to the repression of gene expression, miRNA-mediated upregulation can encompass a spectrum ranging from fine-tuning effects to substantial alterations in the expression of target genes (Vaschetto 2018; Vasudevan 2012). Indeed, dynamic regulations of gene expression by miRNA have been observed in various organisms facing environmental challenges such as thermal stresses, pollutants and pathogen infections in fishes (Sun et al. 2019; Xia et al. 2022; Xu et al. 2019), alkalinity in shrimps (Shi et al. 2022), hypoxia in sea cucumbers (Huo et al. 2017) and desiccation in oysters (Chen et al. 2017). Due to their functional significance, miRNAs have been recognised as sculptors of the transcriptome and cornerstones of gene regulation networks (Solís et al. 2022). Their multidimensional regulations – one miRNA regulates multiple target genes (sometimes even hundreds) while one gene is controlled by several miRNAs – form complex and dynamic regulatory networks (Pu et al. 2019). This complexity provides flexibility and resilience for response to diverse environmental challenges, thereby facilitating phenotypic plasticity crucial for invasion success (Ebert and Sharp 2012; Berezikov 2011; Xiang et al. 2023).

Despite recognising the functional importance of miRNAs in response to varied environments, the systematic, detailed functional process and regulatory mechanism remain largely unexplored, particularly concerning their roles in short-term ecological response. In this study, we utilised the model invasive ascidian *Ciona robusta*, formerly known as *C. intestinalis* sp. A (Caputi et al. 2007), to investigate the causes and consequences of miRNA-mediated regulation of gene expression in response to environmental challenges. As a highly invasive ascidian presumably native to the Northwest Pacific, *C. robusta* has successfully colonised coastal areas globally, including the extreme environmental conditions (e.g., salinity of 40‰–41‰) of the Red Sea (Brunetti et al. 2015; Bouchemousse et al. 2016; Chen et al. 2018; Shenkar et al. 2018). During the invasion process, *C. robusta* frequently encounters and successfully survives recurring acute environmental stresses, such as salinity fluctuations of 15‰ during shipping-mediated range expansions (Briski et al. 2013). As a sessile organism, *C. robusta* cannot actively escape from harsh environments but relies primarily on its robust adaptive homeostasis for survival (Richards et al. 2006). Due to its biological features, high invasiveness, strong adaptive capacity and well-sequenced small genome, *C. robusta* has become a promising model for studying invasion success (Zhan et al. 2015).

Using this model invasive ascidian, we simulated the recurring hyperosmotic environmental challenges that *C. robusta* often encounters during the invasion process and then integrated miRNAome and transcriptome analyses in a controlled experimental system (Figure 1). We aim to (1) explore the expression patterns of miRNAs at different stages of recurring salinity challenges and further identify potential causes for the observed patterns; (2) assess the functional regulatory relationships between miRNAs and target genes and further study possible drivers (e.g., stress memory) for the observed gene expression patterns at different stages of salinity stress and (3) investigate the utilisation and adaptation of canonical osmoregulation strategies in different stages of environmental challenges and further assess the functional diversification

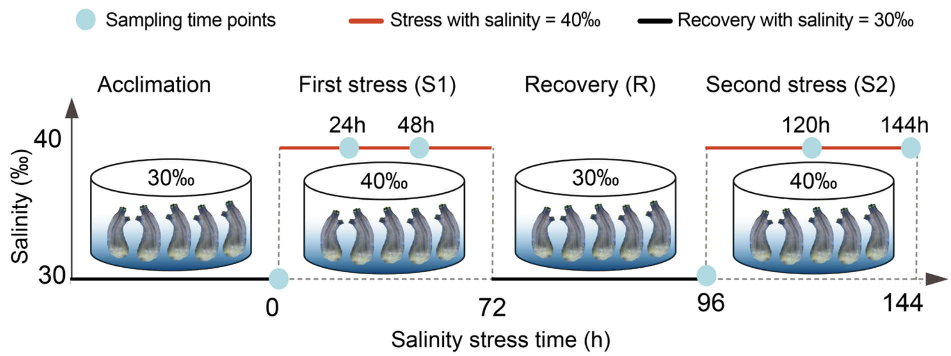

FIGURE 1 | Schematic illustration of the experimental design. After acclimation, healthy *Ciona robusta* individuals were randomly selected for subsequent hyperosmotic stress experiment. *C. robusta* individuals were treated with hyperosmotic stress (40‰) for 72 h (S1 stage) and sampled at 0, 24, 48 h, then transferred into ambient condition (30‰) for 24 h (R stage) and sampled at 96 h, and finally subjected to another round of hyperosmotic stress (40‰) for 72 h (S2 stage) and sampled at 120 and 144 h. [Colour figure can be viewed at wileyonlinelibrary.com]

under miRNA regulation. The dynamic gene expressions and functional diversifications regulated by miRNA obtained in this study are expected to provide a deep understanding of the functional roles of miRNA-mediated networks in short-term ecological responses and facilitate further investigations into the importance of environment-induced plasticity for long-term evolutionary adaptation.

2 | Material and Methods

2.1 | Sample Collection and Recurrent High Salinity Stresses

Ciona robusta adults were collected from the aquaculture facility in Dalian, Liaoning Province, China (38°49′19″ N, 121°2′28″E). During the whole process of the experiment, *C. robusta* individuals were nourished with *Spirulina* sp. and *Chlorella* sp., and healthy individuals were selected for subsequent stress tests in the laboratory. To simulate the environmental challenges faced by *C. robusta* during invasions, we subjected ascidians to recurrent hyperosmotic stresses (Figure 1). The salinity of the control group and the recovery stage were maintained at 30‰, corresponding to the salinity level at the sampling site. The high salinity stress was set to 40‰ by adding sea salt to seawater, which was chosen based on the extreme environments of the Red Sea in the invasive range of this ascidian (Chen et al. 2018). Based on our intensive investigations of gene expression patterns in *Ciona* in response to environmental challenges (e.g., Huang et al. 2017; Huang and Zhan 2021; Huang, Li, Shenkar, and Zhan, 2023; Huang, Li, and Zhan, 2023), we established recurrent salinity challenges as follows and sampled at key time points based on the dynamic expression of both transcriptomic and epigenetic processes: after a 3-day acclimation in aerated natural seawater at 22°C and salinity of 30‰, six individuals were collected for the control group. Subsequently, all other ascidians were transferred to high salinity seawater (40‰) for 72 h (S1 stage), followed by 24 h of recovery (R stage) by relocating individuals to 30‰ seawater. Subsequently, the individuals were re-challenged by high salinity (40‰) for another 72 h (S2 stage). Samples were collected at six time points (Figure 1): 0 h at the end of acclimation (Control), 24 and 48 h at the S1 stage, 96 h at the end of the R stage and 120 h and 144 h at the S2 stage. Somatic muscle tissue from each sample was dissected and immediately frozen in liquid nitrogen for further analyses.

2.2 | RNA Extraction, miRNA and mRNA Library Construction and Sequencing

Total RNA was extracted from the collected samples using Trizol reagent (Ambion, Massachusetts, USA) following the manufacturer's instructions. DNase I (Promega, USA) treatment was performed to eliminate potential genomic DNA contamination. The integrity, purity and quantity of the RNA were assessed using the Agilent 2100 Bioanalyzer, NanoPhotometer and Qubit 3.0 Fluorometer. After the quality control process, RNA molecules with the size ranging from 18 to 30 nt were enriched using polyacrylamide gel electrophoresis (PAGE). Subsequently, 3′ adapters were added and RNAs in the range of 36–48 nt were enriched. Finally, 5′ adapters were ligated to the enriched RNAs, and the ligation products underwent reverse transcription by PCR amplification. PCR products in the size range of 140–160 bp were enriched to create a miRNA library, and then, the constructed library was sequenced on the Illumina Novaseq 6000 platform using the PE150 strategy. The same samples were parallelly utilised for transcriptomic analyses. The RNA-sequencing libraries were prepared according to the instructions of the NEBNext Ultra RNA Library Prep Kit (New England Biolabs, Massachusetts, USA) and then sequenced on the Illumina HiSeqX Ten Sequencing System (Illumina, California, USA) with the PE150 strategy. The transcriptomic and miRNAomic data were deposited in the National Centre for Biotechnology Information (NCBI) Sequence Read Archive (SRA) database under the accession number PRJNA775866 and PRJNA1120970, respectively.

2.3 | miRNA Identification and Differential Expression Analysis

After sequencing, we eliminated low quality reads that contained more than one base with *Q*-value ≤ 20 and with unknown nucleotides (N) and then removed adapters from raw reads using Cutadapt v4.5 (Martin 2011). The clean tags were mapped to the reference genome of *C. robusta* (HT version, http://ghost.zool.kyoto-u.ac.jp/default_ht.html) using Bowtie2 v2.5.1 (Langmead and Salzberg 2012). Subsequently, miRDeep2 v0.1.3 was employed to identify and quantify miRNAs based on read counts (Friedländer et al. 2012). The expression levels of miRNAs were calculated and normalised from read counts to transcripts per million (TPM), and the miRNAs with TPM > 0 in more than 50% of the samples were retained for subsequent analyses. To evaluate the differential expression of miRNAs induced by recurrent osmotic stresses, the 'DEseq2' package was applied for identifying differentially expressed miRNAs (DEmiRNAs) between control stage (0 h) and other sampling points (24, 48, 96, 120, 144 h) as well as between R stage (96 h) and S2 stage (120 and 144 h), and the threshold for DEmiRNAs was *p. adj* < 0.05 and $|\log_2 \text{foldchange}| > 1$ (Love et al. 2014). To visualise the overall miRNA expression differences among all samples, we conducted principal component analysis (PCA) using *procmp* in R. We used the 'pheatmap' package and 'ggplot2' package to generate a heatmap illustrating the expression profiles of DEmiRNAs (Kolde 2019; Wilkinson 2011). We used the Venn diagram tool on the OmicShare platform (www.omicshare.com/tools) to identify overlapping and unique DEmiRNAs among stages.

2.4 | Identification and Annotation of the Target mRNAs of DEmiRNAs

To assess the biological functions of DEmiRNAs, we employed two algorithms, RNAhybrid and miRanda (http://www.microrna.org/), to predict the target mRNAs for DEmiRNAs (Betel et al. 2008; Rehmsmeier et al. 2004). The prediction threshold was set at an energy level of ≤ -20 kcal/mol, and the overlapping results of these two algorithms were utilised for further analysis.

For the transcriptomic analysis, clean mRNA sequencing reads were initially mapped to the reference genome of *C. robusta* using Hisat2 v2.1.0 (Kim et al. 2019). Subsequently, we employed featureCounts to obtain read counts of genes and 'DESeq2' package to identify differentially expressed genes (Liao et al. 2013; Love et al. 2014). If the predicted target genes of DEmiRNAs also exhibited significant differential expression under recurrent high salinity stresses with a threshold of *p. adj* < 0.05 and $|\log_2 \text{foldchange}| > 1$, they were identified as differentially expressed target genes (DETGs). We conducted the Gene Ontology (GO) enrichment analysis for DETGs at various sampling time points and stages using the Omicshare tools (https://www.omicshare.com/tools). Significantly enriched GO terms (*p. adj* < 0.05) were selected as representations of functional categories (Ashburner et al. 2000).

For the differential miRNA and mRNA response patterns between the two rounds of hyperosmotic stress, as we did not detect significant type I stress memory, we only analysed type II stress memory as defined by Oberkofler et al. (2021). This refers to an enhanced response upon the second stress exposure compared to the first. According to Oberkofler et al. (2021), we define the enhanced downregulation at the second round of stress as type II–a stress memory, while the enhanced upregulation at the second round of stress as type II–b stress memory. Besides, the function of DETGs which showed type II stress memory was analysed utilising GO enrichment analysis, and the expression patterns of DEmiRNAs and corresponding DETGs were visualised using 'pheatmap' package in R (Kolde 2019).

2.5 | Expression Correlation of miRNA and Target mRNA

To infer the regulatory relationship between miRNA and target mRNA, we calculated the Pearson correlation coefficient for the expression of miRNAs and their respective mRNA targets across all samples using the 'psych' package in R (Revelle 2019). We set a strong correlation threshold at $|\text{correlation coefficient}| > 0.8$ and considered correlations with *p. adj* < 0.05 as significant. Additionally, we analysed the distribution of binding sites in target genes using the 'GenomicRanges' package in R (Lawrence et al. 2013).

2.6 | miRNA-Mediated Canonical Osmotic Stress Response Genes

Genes related to the three canonical strategies of cellular osmotic homeostasis, including ion channel, free amino acid (FAA) metabolism and biosynthesis, water channel, were retrieved from the annotation of *C. robusta* reference genome. These canonical osmotic stress response genes were selected based on previous research in aquatic animals (Cao and Shi 2019; Liu et al. 2019; Lou et al. 2019; Pu and Zhan 2017; Towle et al. 2011; Verri et al. 2012). Next, we constructed a regulatory network between the targets of the three strategies and their related miRNAs using the 'circlize' package in R (Gu et al. 2014). Finally, we performed Gene Set Enrichment Analysis (GSEA) using 53 genes related to ion transport and 31 genes (see the 3. Results section) related to FAA metabolism and biosynthesis to investigate their expression profiles, with the thresholds of *p. adj* < 0.05, $|\text{Normalised Enrichment Score (NES)}| > 1$ and *FDR* < 0.25 (Subramanian et al. 2005).

2.7 | Effects of Key miRNA Biogenesis- and Decay-Related Genes on miRNA Abundance

The variation of overall miRNA abundance ($\log_{10}$TPM) among different sampling points was investigated through the Wilcoxon test and visualised in a boxplot. After that, to evaluate the potential causes of expression alteration of miRNAs, the key genes related to the canonical miRNA biogenesis and degradation pathway were identified according to previous studies (De Bofill-Ros and Vang Ørom 2024; Han and Mendell 2023; Shang et al. 2023; Treiber et al. 2019). The expression patterns of these genes were extracted from the transcriptome data, retaining only the differentially expressed genes related to functional biogenesis and degradation for further analyses. The correlation relationship between the differentially expressed biogenesis/degradation genes and global miRNA abundance was analysed using the mantel test through the vegan package in R (Oberkofler et al. 2021). Furthermore, to investigate which specific miRNAs had a correlation relationship with biogenesis/degradation genes, iNAP (integrated Network Analysis Pipeline) was utilised for constructing an integrated network and visualisation was

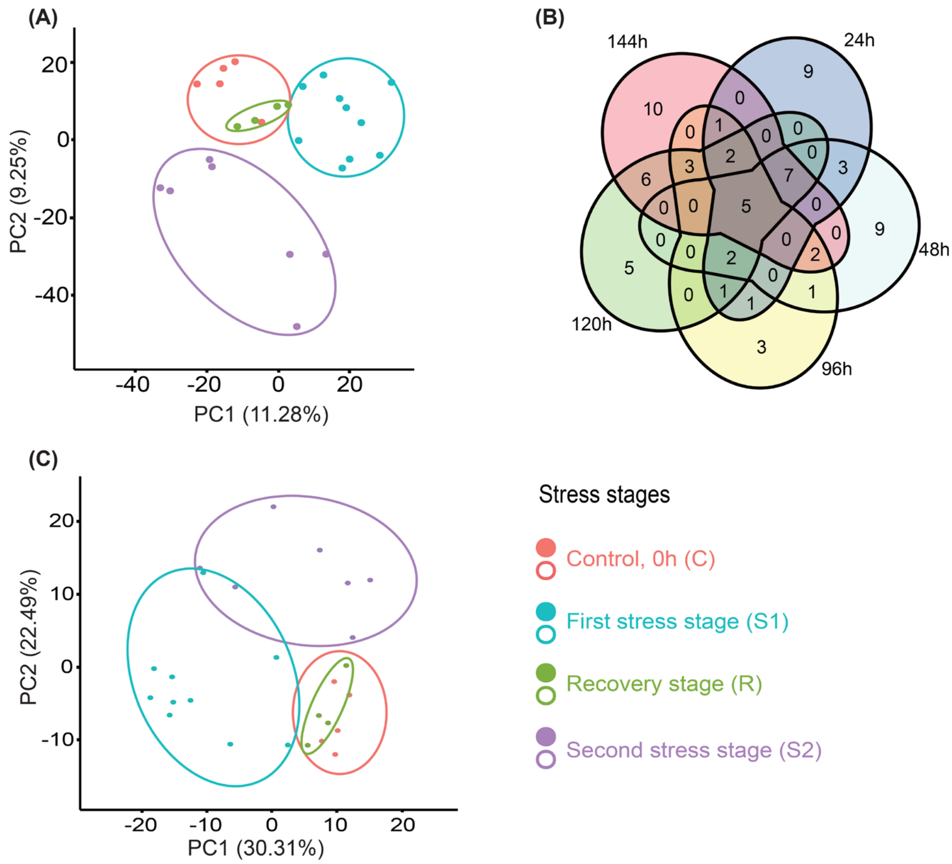

FIGURE 2 | (A) The principal component analysis (PCA) plot showing the differences and similarities in all miRNA expression patterns in the control group (C, red), first stress stage (S1, blue), recovery stage (R, green), second stress stage (S2, purple) along the principal components 1 (PC1) and 2 (PC2). The percentage of total variation explained by each principal component axis is given in parentheses. (B) The Venn diagram of differentially expressed miRNAs (DEmiRNAs) in different sampling time points, including 24 h (blue), 48 h (light green), 96 h (yellow), 120 h (dark green) and 144 h (red). (C) The PCA plot of DEmiRNAs during recurrent salinity stresses, and four stages were labelled in red (C), blue (S1), green (R) and purple (S2). [Colour figure can be viewed at wileyonlinelibrary.com]

conducted by the online OmicShare tools (https://www.omicshare.com/tools) (Feng et al. 2022).

3 | Results

3.1 | Dynamic miRNA Expression Response

After sequencing and quality control, we obtained an average of 1.7 million clean reads per sample, with approximately 75% of the reads mapped to the reference genome (Table S1). From the sequence data, a total of 889 miRNAs were identified, including 197 known and 692 novel miRNAs after classification. The length of miRNAs ranged from 18 to 25 nucleotides, with the highest peak at 22 nucleotides (Figure S1A). Nucleotide base analysis revealed that uracil (U) was the dominant base at the first position (Figure S1B). These conserved sequence features of miRNAs indicate high sequence quality and appropriate data processing for subsequent analyses.

PCA analysis was performed to explore the similarities and differences in miRNA expression profiles across different stress stages (Figure 2). The first two principal components (PC1 and PC2) explained 20.53% of the total variation (Figure 2A), and the PCA results revealed distinct clustering patterns: samples from the S1 and S2 stages formed two separate clusters, exhibiting significant differences from the control stage. In contrast, the control stage and the recovery (R) stage did not form distinct clusters, suggesting that miRNA expression patterns at the recovery stage returned to levels similar to those of the control (Figure 2A). These intriguing patterns illustrate that, despite exposure to the same stress, repeated challenges elicited diverse miRNA responses in terms of duration.

When compared with the control, we identified 31, 29, 21, 31 and 36 DEmiRNAs at 24, 48, 96, 120 and 144 h, respectively. Additionally, 17 and 15 miRNAs showed differential expression at 120 and 144 h (the 2nd round of salinity stress) when compared with 96 h (the recovery stage between two rounds of challenges). This resulted in a total of 70 DEmiRNAs across all time points. Among these, 36 DEmiRNAs (51.43%) exhibited significant differential expression at only one sampling time point. Furthermore, 11, 16 and two DEmiRNAs responded to the salinity stress at two, three and four time points, respectively. Notably, only five miRNAs exhibited differential expressions across all five sampling points (Figure 2B). This pattern underscores the dynamic miRNA expression response to salinity stresses, with different miRNAs being involved at different stress stages.

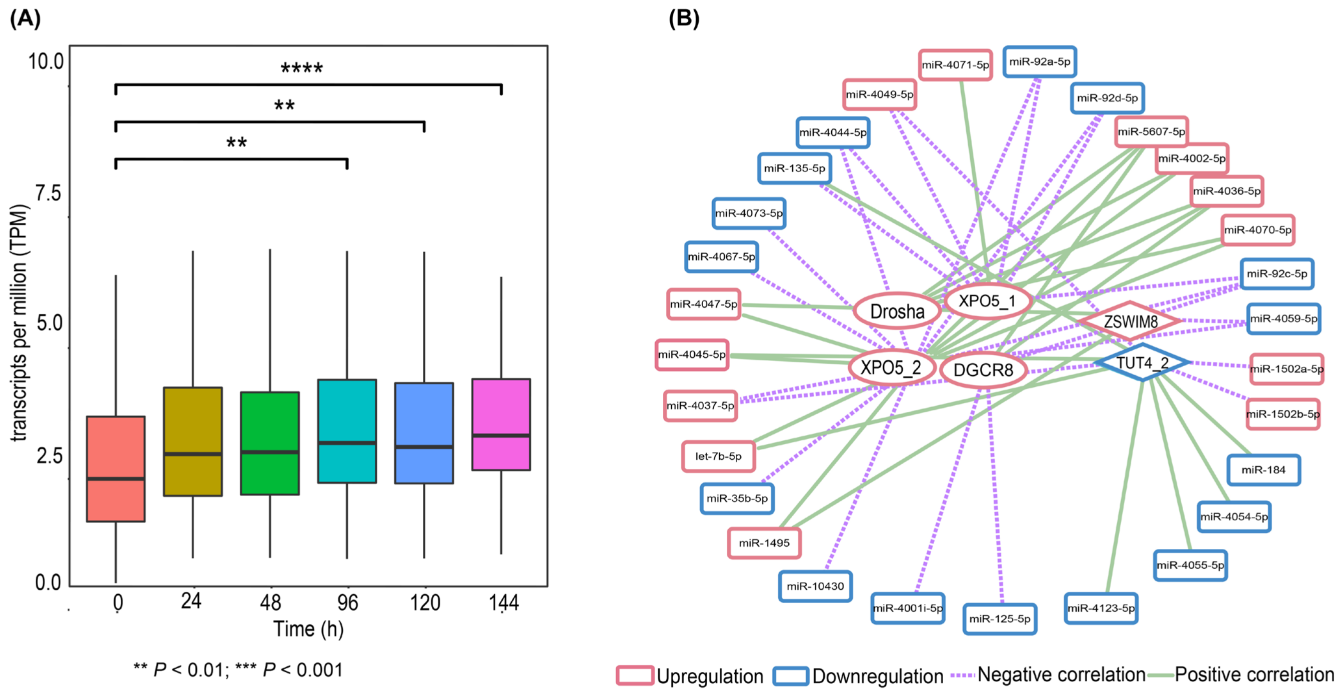

FIGURE 3 | (A) The global abundance of miRNAs in different sampling points (24, 48, 96, 120 and 144 h). The difference of $\log_{10}$TPM between 0 h and other sampling points was estimated using the Wilcoxon test ($p < 0.05$). (B) The direct regulation relationships between the miRNA biogenesis/decay genes and differentially expressed miRNAs (DEmiRNAs). The biogenesis genes, decay genes and DEmiRNAs are shown in ellipse, rhombus and rounded rectangles, respectively. Up and downregulations are illustrated in red and blue, and the dotted and full lines were used to show negative and positive regulation relationships, respectively. [Colour figure can be viewed at wileyonlinelibrary.com]

The PCA plot based on the expression of 70 DEmiRNAs revealed that the first two principal components explained 52.8% of the total variation, which is higher than that obtained from the PCA analysis based on all miRNAs (Figure 2C). This indicates that the DEmiRNAs were the primary contributors to differential expression during the two rounds of osmotic stress. The PCA results demonstrated clear differentiation between S1 and control stages, S2 and control stages, as well as between S1 and S2 stages. Furthermore, the R stage did not separate from the control stage, illustrating that there was no significant dissimilarity in DEmiRNA expression patterns between the control and R stages (Figure 2C). This finding highlights the duration- and stage-specific expression of miRNAs in response to osmotic stresses.

3.2 | Major Biological Processes Driving Dynamic miRNA Expression Response

To delve deeper into the major biological processes driving the observed dynamic miRNA expression response patterns, we focused on two primary processes: miRNA biogenesis and decay and further assessed the known genes of these two processes on miRNA expressions. We found that the overall miRNA abundance at 96, 120 and 144 h significantly increased compared to the control stage (Figure 3A). To elucidate the underlying drivers for this pattern, we identified key genes related to miRNA biogenesis and decay from the genome of *C. robusta* (Table S3). A Mantel test assessing the effects of these genes on overall miRNA expression showed that three differentially expressed biogenesis genes (*DGCR8*, *Drosha*, *XPO5_2*) had a significant positive correlation with global miRNA abundance (Figure S2). In contrast, three differentially expressed decay genes (*ZSWIM8*, *TUT4_1*, *TUT4_2*) had no significant influence on global miRNA abundance. These patterns indicate that the canonical biogenesis pathway, rather than degradation, should be the main driver affecting miRNA abundance in response to salinity stresses.

To further evaluate which miRNAs were influenced by these genes, the integrated network analysis identified that four biogenesis genes (*DGCR8*, *Drosha*, *XPO5_1*, *XPO5_2*) and two decay genes (*ZSWIM8*, *TUT4_2*) had significant effects on 23 DEmiRNAs (Figure 3B). A total of 10 DEmiRNAs that were negatively correlated with biogenesis genes were downregulated after stress, while five DEmiRNAs positively correlated with biogenesis genes were upregulated, suggesting that an enhanced biogenesis process should elevate miRNA abundance and vice versa. Additionally, two DEmiRNAs negatively regulated by TUT4_2 were downregulated, whereas four DEmiRNAs positively regulated by TUT4_2 were upregulated. The expression patterns of DEmiRNAs influenced simultaneously by biogenesis and decay genes varied, indicating that the complex mechanisms underlying their expression alterations require further detailed assessment.

3.3 | Functional Regulatory Relationships Between miRNAs and Target Gene

A total of 4434 target genes were identified for the 70 DEmiRNAs, with the number of target genes for each miRNA ranging from

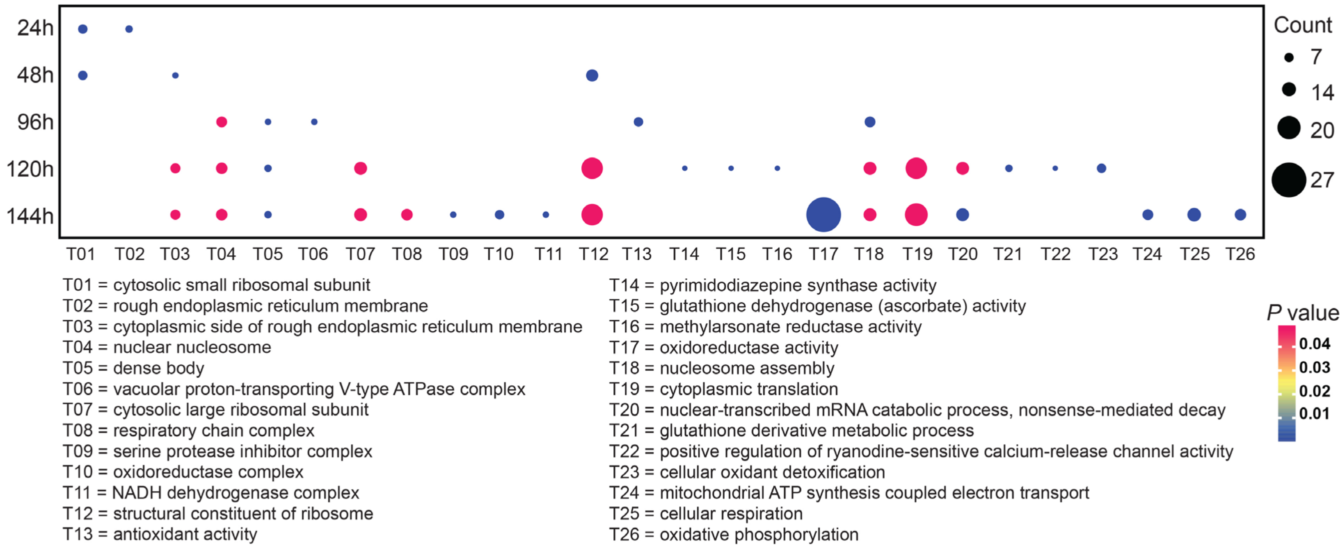

FIGURE 4 | The bubble chart of 26 enriched Gene Ontology (GO) terms in five sampling points (24, 48, 96, 120 and 144 h) compared to the baseline at 0 h. The node size and colour depth of bubbles represent the number of targets involved and *p.adj* of every term. [Colour figure can be viewed at wileyonlinelibrary.com]

seven to 742, and an average of 131. Among these target genes, 2106 (47.50%) were identified as DETGs in the transcriptomic data. Due to the multiple-to-multiple regulation relationships between miRNAs and their target genes, we detected a total of 9199 miRNA-target gene pairs.

Gene Ontology enrichment analysis was conducted to elucidate the functional representations of DETGs at different sampling points and among stages. At 24 and 48 h during the first stress, the functions of the target genes were primarily enriched in cellular component terms, while enriched terms for target genes at 96 h were related to the cell nucleus, antioxidant activity and proton transport. In contrast, the target genes at 120 h were associated with enzymatic activity, translation, mRNA catabolic processes, ion channels and cellular oxidant detoxification. The target genes at 144 h showed enrichment in terms related to the respiratory chain complex, translation, mRNA catabolic processes, mitochondrial ATP synthesis coupled with electron transport, cellular respiration and oxidative phosphorylation (Figure 4). DETGs in the second stress stage (S2) showed a greater enrichment of GO terms compared to those in the first stress stage (S1; Figure S3). The enriched terms in S1 were primarily associated with the mitochondrial matrix and methyltransferase activity. In contrast, some stress response-related terms, such as antioxidant activity and ion channels, were specifically observed in S2. Overall, the functions of differentially expressed target genes under the regulation of differentially expressed miRNAs were diverse and distinct at different sampling points in the treatment stages.

Among the 9199 miRNA-target pairs identified, approximately 45% exhibited positive expression correlations, while the remaining 55% displayed negative correlations (Figure S4). Further analysis revealed 179 (1.9%) significantly correlated miRNA-mRNA relationships, with approximately 70% of them being positively correlated. Among these, 32 target pairs showed strong correlations with correlation coefficients exceeding 0.8.

Regarding the distribution of binding sites in target genes, we observed that coding sequences (CDS) represented the most prevalent one, accounting for approximately 89% of the binding sites, while the 5′UTR and 3′UTR accounted for 8% and 3% of the binding sites, respectively (Figure S5). Interestingly, we did not observe explicit effects of the binding site location on the miRNA-mRNA regulatory relationship, as binding sites for both positive and negative regulatory relationships were found in the 5′UTR, CDS and 3′UTR (Figure S6).

3.4 | miRNA-Mediated Formation of Stress Memory of Target Genes

We observed stress memory in both DEmiRNAs and their corresponding DETGs (Figure 5). Specifically, among DEmiRNAs, a total of 12 and 11 miRNAs exhibited type II-a (Figure 4A) and type II-b stress memory (Figure 5B), respectively. No significant type I stress memory was detected in this study. Interestingly, the corresponding DETGs of type II-a DEmiRNAs were categorised into both type II-b (Figure 5C) and type II-a stress memory (Figure 5D). Type II-b DETGs were associated with the regulation of mRNA stability, while type II-a DETGs were involved in selenoamino acid/tryptophan metabolism, metal ion transport and potassium (K^+) channels (Figure 5G). Similarly, the corresponding DETGs of type II-b DEmiRNAs were also classified into both types (Figure 5E,F). Type II-a DETGs included functions related to alanine and aspartate metabolism, K^+/Ca^{2+} channels, and the mitochondrial respiratory chain, whereas type II-b DETGs were associated with Ca^{2+} channels, regulation of mRNA stability and tryptophan metabolism (Figure 5H). Notably, type II-a DEmiRNAs positively regulated type II-a DETGs and negatively regulated type II-b DETGs, while type II-b DEmiRNAs positively regulated type II-b DETGs and negatively regulated type II-a DETGs. In summary, type II stress memory DEmiRNAs regulated DETGs that exhibited stress memory, illustrating the role of DEmiRNAs in the formation of mRNA-level stress memory.

3.5 | miRNA-Mediated Canonical Osmotic Stress Coping Strategies

A total of 92 canonical osmotic stress response genes were identified from the miRNA regulatory target gene list. When categorising into three osmotic stress coping strategies, 53 genes were related to ion transport, 36 to free amino acid (FAA) metabolism and biosynthesis, and only three were associated with water transport (Table S4). The majority of these genes, excluding three aquaporin genes, showed notable differential expression during recurrent osmotic stresses. Our Gene Set Enrichment Analysis (GSEA) results revealed a significant downregulation of the ion transport process at 120 and 144 h (Figure 6A,B), while the FAA metabolism and biosynthesis process was downregulated at 96 h (Figure 6C). Of particular interest, the top core enrichment genes associated with ion transport at 120 h included solute carrier family 39 member 1 (*SLC39A1*) (metal ion), which was negatively regulated by cin-miR-4037-5p. Similarly, core enrichment genes related to ion transport at 144 h included calcium voltage-gated channel auxiliary subunit gamma 3 (*CACNG3*) (calcium transport), which was negatively regulated by cin-miR-4074-5p. Furthermore, the top core enrichment genes involved in the FAA metabolism and biosynthesis at 96 h included cytochrome P450 family 27 subfamily B member 1 (*CYP27B1*), which was positively regulated by cin-miR-135-5p.

The regulatory relationships between these miRNAs and their corresponding mRNA targets revealed the involvement of DEmiRNAs in the regulation of osmotic stress response genes (Figure 7). Interestingly, we observed miRNA-mediated functional strategy transformation from S1 stage to R stage and then to S2 stage, although some functions of DETGs overlapped (e.g., ion transport). Specifically, at the S1 stage, DEmiRNAs regulated K^+ channel (*KCNK2*, *HCN1*, *KCNH3*), Ca^{2+} channel (*CACNA1A*), ion channel regulator and biogenesis of FAAs. In comparison, DEmiRNAs at the R stage regulated the H^+ transport process

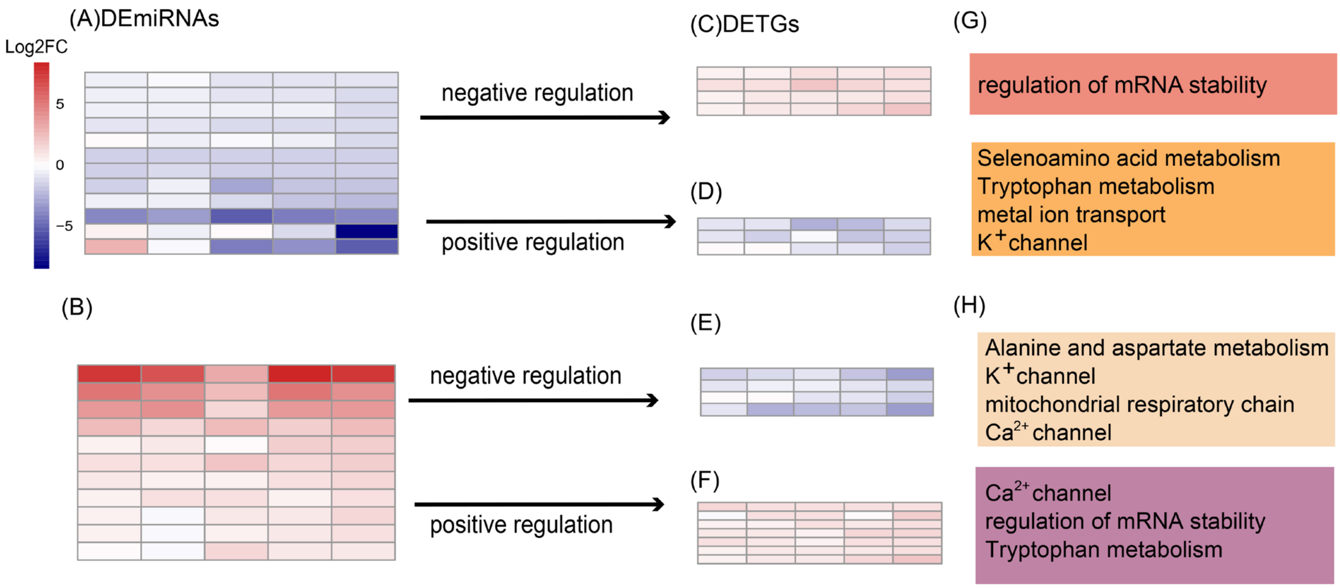

FIGURE 5 | (A, B) The heatmaps of 'memory' differentially expressed miRNAs (DEmiRNAs) that showed enhanced response patterns in the second stress (S2) stage; (A) represents type II-a memory DEmiRNAs while (B) represents type II-b memory DEmiRNAs. (C, D) The heatmaps of differentially expressed target genes (DETGs) of type II-a DEmiRNAs which showed two 'memory' patterns. (E, F) The heatmaps of DETGs of type II-b DEmiRNAs which showed two 'memory' patterns. (G, H) The functions of DETGs are shown in the right side of heatmaps. [Colour figure can be viewed at wileyonlinelibrary.com]

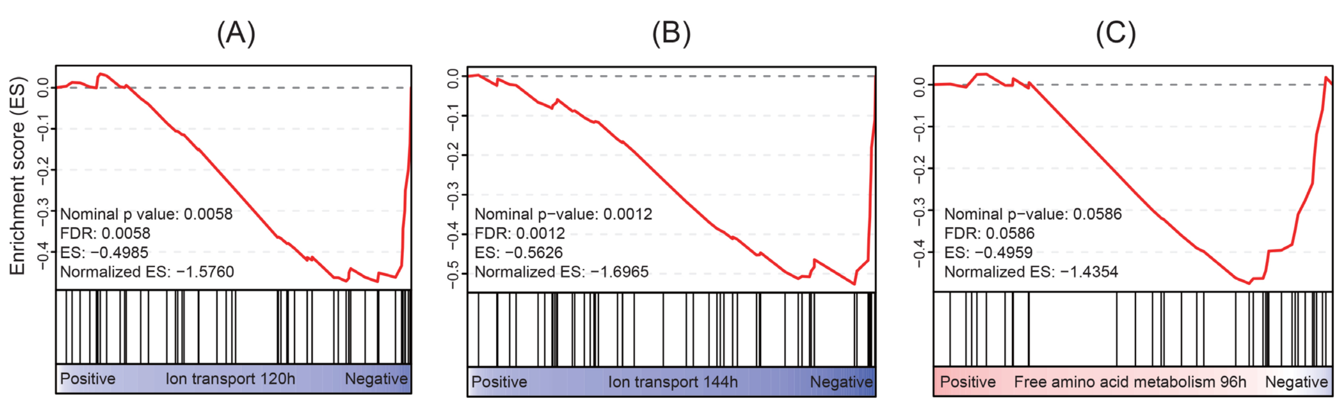

FIGURE 6 | The Gene Set Enrichment Analysis (GSEA) results showing significant enrichment of two gene sets, including downregulation of ion transport gene set at 120 h (as shown in Figure 4A) and 144 h (as shown in Figure 4B), as well as downregulation of free amino acid metabolism gene set at 96 h (as shown in Figure 4C). [Colour figure can be viewed at wileyonlinelibrary.com]

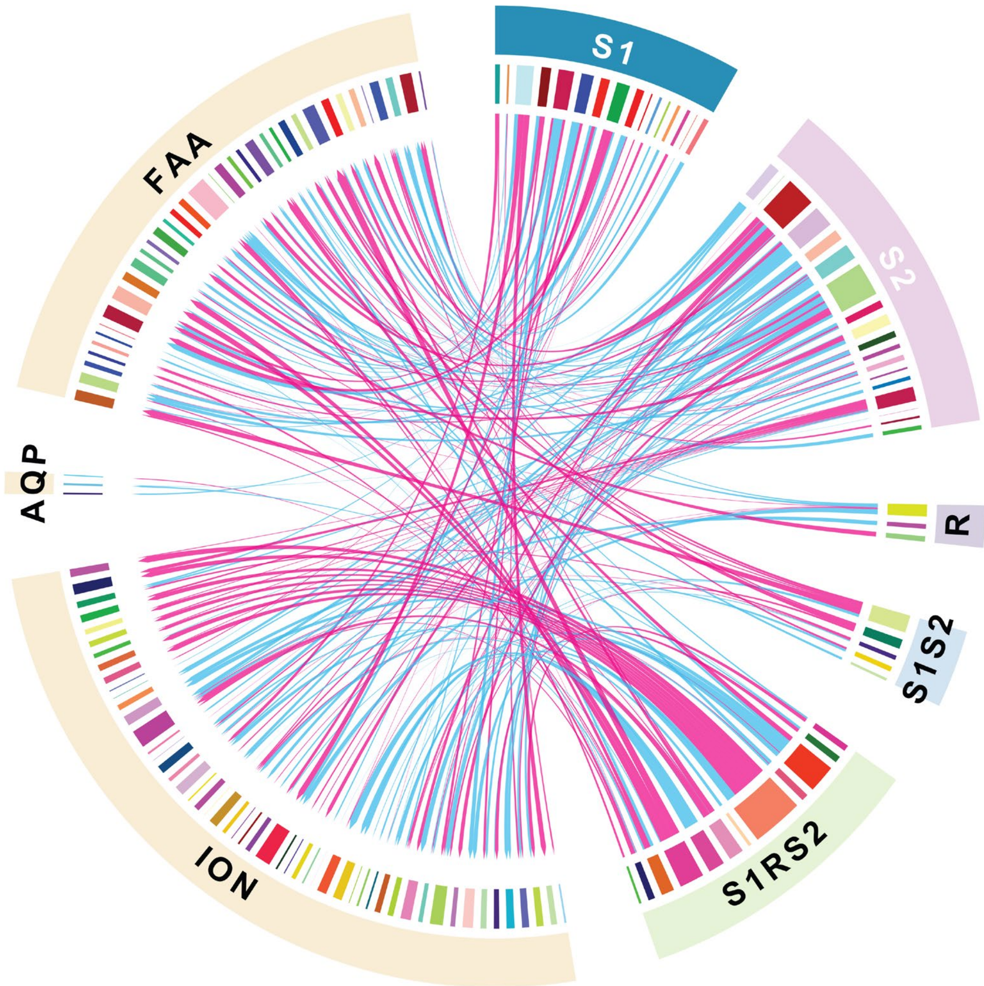

FIGURE 7 | The regulation network of differentially expressed miRNAs (DEmiRNAs) and functional targets. The targets (left side) were divided into three group according to their function – FAA = free amino acid (FAA) metabolism and biosynthesis, ION = ion transport, and AQP = aquaporin (water transport). The DEmiRNAs (right side) were classified based on their respond stages, 'S1', 'R' and 'S2' represents responsive miRNAs only in the first stress stage (S1), recovery stage (R) and second stress stage (S2), respectively, while 'S1S2' represents responsive miRNAs in both S1 and R and 'S1RS2' represents responsive miRNAs in S1, R and S2. The link between miRNAs and corresponding targets shows their regulation patterns, red line represents positive correlation while blue line represents negative correlation. The width of link shows the correlation strength. [Colour figure can be viewed at wileyonlinelibrary.com]

(*ATP6AP1L*), and those at the S2 stage regulated K^+ channel (*KCNE5*, *KCNK5*, *KCNK16*, *KCNS1*), Ca^{2+} channel (*CACNA1H*, *CACNA1I*, *TRPA1*), Cl^- channel, Zn^{2+} transport, organic cation/anion/zwitterion transport, H^+ transport process (*SLC9B1P1*), ion channel regulator, as well as FAA metabolism (glutathione, tryptophan, valine, leucine, isoleucine, phenylalanine). Regarding the miRNA-mRNA regulatory relationship, there was a slight increase in negative regulations: 53% vs. 47%, 50% vs. 50% and 42% vs. 58% for positive and negative regulations at S1, R and S2 stages, respectively. These findings indicate that differentially expressed miRNAs regulate various strategies and associated functional groups of genes at different stages to respond to the same environmental stress, showcasing a diversification of the dynamic regulatory network in response to the osmotic stress.

4 | Discussion

Understanding miRNA-mediated regulatory networks is crucial for elucidating the formation of short-term phenotypic plasticity, integrating environment-induced variations into long-term evolutionary adaptation, and finally uncovering coping strategies and associated mechanisms that determine whether organisms survive or succumb to rapid environmental changes. Biological invasions provide a promising system to explore these mechanisms, as invaders must employ effective strategies to cope with rapid environmental changes during invasions. As a result, rapid responses to environmental stresses are crucial for invasion success, as they initially contribute to the survival of invaders and the maintenance of population size. Within complex regulatory networks, miRNAs can functionally tune the expression of coding genes, thus facilitating organisms' rapid responses to environmental challenges. Using the model invasive ascidian, *C. robusta*, here we simulated recurrent salinity challenges during invasions and investigated the causes and consequences of miRNA-mediated regulations on gene expression. Our findings highlight the rapid and dynamic miRNA expression responses to salinity stresses with duration-dependent and stage-specific miRNA profiles (Table S2, Figure 2A). Overall, the upregulation of the canonical biogenesis process, rather than the decay pathway, contributed more to the increased global abundance of miRNAs in response to recurrent salinity stresses (Figure 3 and Figure S2). Interestingly, we detected miRNA-mediated formation of stress memory in corresponding target genes, which exhibited more pronounced reactions during the second stress stage in comparison to their responses in the initial stage (Figure 4). Functionally, the biological processes regulated by DEmiRNAs exhibited temporal variations, primarily encompassing the modulation of FAA metabolism and ion transport strategies to maintain osmotic homeostasis under ambient salinity fluctuations (Figures 6 and 7 and Figure S3). Furthermore, our results revealed diverse miRNA-mRNA regulatory patterns, demonstrated by both the interaction relationships (Figure S4) and binding sites (Figure S5). These findings underscore the intricate, dynamic and effective nature of regulatory networks, emphasising the diversified miRNA-mediated regulations on gene expression in response to environmental challenges.

4.1 | Causes and Consequences of Dynamic miRNA Expression in Response to Recurrent Stresses

The use of epigenetic processes enables organisms to swiftly and flexibly respond to environmental changes (e.g., Geng et al. 2012). In this study, we observed rapid miRNA responses to salinity stress, with 31 out of 70 DEmiRNAs significantly changing their expression levels as soon as 24h after hyperosmotic stress (Table S2). This represents a relatively fast response speed at the miRNA level as documented in published literature. For instance, the expression of miR-429 in the gill of *Odontesthes humensis* significantly changed in response to osmotic stresses within 24h (Pagano et al. 2022). In contrast, many other studies have observed much longer response times, such as the differential expression of miRNAs arising after 7days of salinity change in *Oncorhynchus mykiss* (Zhou et al. 2022). During the invasion process of *C. robusta*, rapid responses aid in buffering the influence of sudden external stimuli, increasing the likelihood of survival and contributing to invasion success (Angers et al. 2010).

Along the time course of repeated hyperosmotic stresses, different miRNAs were dynamically induced at different stress durations, particularly at different stages of stress treatments (Figure 2B and Table S2). This pattern, which was also documented in several other species (e.g., de Freitas Gues et al. 2018), illustrates that the previous experience of environmental challenges has an influence on subsequent miRNA-mediated response. Besides the dynamic response pattern, the biological functions regulated by miRNAs also showed a stage-specific and time-independent pattern, hinting at a regulatory transformation on functions along with timescales (Figure 4 and Figure S3). For example, the biological functions of miRNA-mediated DETGs transferred from methyltransferase activity at the S1 stage to antioxidant activity at the S2 stage (Figure S3). The former miRNA-mediated change likely functions by protecting membrane integrity, while the latter counteracts the oxidative stress triggered by salinity changes (Kushawaha et al. 2021; Liu et al. 2015; Sharma et al. 2012).

In addition to the consequences of dynamic miRNA expression in response to stresses, we further elucidated the causes of miRNA abundance alterations by investigating changes in miRNA biogenesis and degradation processes (Figure 3B and Figure S2). Studies have shown that environmental stresses can influence the gene expression involved in miRNA biogenesis and degradation processes (Cui et al. 2016; Ng et al. 2020). Similar to those studies, we found differential expressions of six biogenesis genes (*Drosha*, *XPO1*, *XPO5_1*, *XPO5_2*, *XPO4_3*, *DGCR8*) and three decay-related genes (*ZSWIM8_2*, *TUT4_1*, *TUT4_2*) under the osmotic stress (Table S3), alongside significant changes in global miRNA abundance (Figure 3A). Studies have discovered causal relationships between changes in biogenesis genes and overall miRNA abundance (Kim et al. 2016; Sun et al. 2016). For instance, the knockout of the *Drosha* gene reduced miRNA levels produced via the canonical biogenesis pathway (Kim et al. 2016), while *XPO5* phosphorylation inhibited pre-miRNA export, causing global downregulation of mature miRNAs (Sun et al. 2016). Indeed, the global miRNA abundance change is a balance between miRNA biogenesis and degradation processes. However, our results demonstrated a significant positive correlation between three upregulated biogenesis genes (*Drosha*, *DGCR8*, *XPO5_2*) and global miRNA abundance (Figure S2), while no direct correlation was observed between decay genes and miRNA abundance (Figure 3A). This pattern indicates that the osmotic stress triggered the enhanced biogenesis process but had limited influence on the degradation process, resulting in the observed elevated miRNA levels at 96, 120 and 144h. However, the influence of miRNA abundance alterations cannot be overlooked. An increase in miRNA abundance indicates pronounced regulation of corresponding targets, which, in turn, affects the global abundance of proteins translated from these target genes. From a cellular biology perspective, this regulation also impacts cellular activity and energy production mediated by these targets (Khalid et al. 2014).

4.2 | miRNA-Mediated Formation of Stress Memory

Remembering previous environmental stresses can help individuals respond more effectively or quickly to subsequent exposures, allowing them to allocate more energy to other fitness-related functions or plasticity (van Hulten et al. 2006). In this study, we detected the type II stress memory for some miRNAs, characterised by an enhanced response during the second round of stress compared to the first (Figure 5A,E). Similarly, in *Arabidopsis*, the expression levels of miR824 gradually increased with each treatment over three cycles of heat stresses, with this enhancement altering the developmental process by regulating transcription factors (Szaker et al. 2019). Interestingly, we found that the formation of miRNA stress memory mediated the development of stress memory in their target genes (Figure 3). This pattern suggests that miRNAs should act as potential factors facilitating the memory formation of gene expression. Despite largely unknown mechanisms underlying stress memory, several possible mechanisms have been identified, including sustained expression alterations of transcription factors (Crisp et al. 2016). For instance, miR156 was found to mediate the heat shock memory by regulating the critical transcription factor squamosa-promoter binding-like (SPL)—the overexpression of miR156 enhanced the heat shock memory while its depletion compromised it (Stief et al. 2014).

Beyond transcription factors, chromatin remodelling—encompassing DNA methylation, histone tail modifications and RNA Pol II pausing—also plays a crucial role in stress memory formation. In brief, stress-responsive genes exist in two transcriptional states: active ('switched on') or inactive ('turned off'). Chromatin remodelling mediates transitions between these states, leaving behind gene expression records known as 'memory' arks, which can influence future gene expression patterns (Berger 2007; Bruce et al. 2007). For instance, di-methylation of H3-K4 marks a 'permissive' transcriptional state, whereas tri-methylation is associated with active transcription (Santos-Rosa et al. 2002; Bruce et al. 2007). Additionally, stress memory may also be reinforced through the stabilisation of stress-responsive miRNAs within the RNA-induced silencing complex (RISC), enabling more rapid post-transcriptional regulation of target genes upon subsequent stress exposures (Chendrimada et al. 2005). Despite these possible mechanisms underlying stress memory, we did not detect any evidence that miRNAs mediate mRNA stress memory via the transcription factor regulation in this study. Thus, the molecular mechanisms by which miRNAs contribute to mRNA stress memory are diverse, complex and likely species-specific.

4.3 | miRNA-Triggered Canonical Osmoregulatory

Aquatic animals recruit organic osmolytes, such as free amino acids (FAAs), and inorganic osmolytes, including ions, to cope with salinity fluctuations. The specific osmolytes regulated by miRNAs exhibit a species-specific pattern (e.g., Meng et al. 2013). Indeed, the specific osmolytes regulated by miRNAs exhibit a species-specific pattern (Ibrahim et al. 2023; Shwe et al. 2022; Zhou et al. 2022). As osmoconformers, ascidians have the ability to adjust their intracellular osmolarity to cope with abrupt changes in salinity (Sokolov and Sokolova 2019). Our results showed that ions were the preferential osmolytes to regulate the osmolarity at the miRNA level when exposed to hyper-salinity stress (Figure 6A,B). Previous studies have found that miR-1788-3p and miR-11987-5p regulate *NKAα1* and *NKAα2*, respectively, in *Rachycentron canadum* exposed to hypo-salinity stresses (Huang et al. 2021). Besides, miRNAs have been shown to regulate calcium ion binding and transport activity during salinity acclimation in juvenile *Oncorhynchus mykiss* (Zhou et al. 2022). Apart from regulating ion transport processes, miRNAs have been shown to modulate FAA metabolism, contributing to the maintenance of intracellular osmolality and cell volume in *Crassostrea hongkongensis* (Ibrahim et al. 2023). These findings suggest a conserved functional role of miRNAs in the osmotic stress response across aquatic animals. Interestingly, previous analysis showed that FAAs were the preferred osmolytes under salinity stress to maintain osmotic homeostasis at the transcriptomic level in *C. savignyi*, another highly invasive ascidian in the genus of *Ciona* (Huang, Li, Shenkar, and Zhan, 2023). This pattern suggests species-specific osmoregulation strategies at various regulatory levels. Collectively, miRNAs could regulate different genes related to FAA metabolism and ion transport to alter intracellular osmolarity, and distinct biological processes regulate complementary targets to achieve nonredundant roles when experiencing salinity fluctuations.

4.4 | miRNA-mRNA Regulations: Diverse Binding Sites and Regulatory Relationships

It is well-known that miRNA can bind to the 3′UTR of target genes and suppress their expression (Bartel 2009). Recently, emerging evidence has shown that regulatory mechanisms beyond the well accepted canonical pathway exist. MiRNAs can bind diverse genomic regions such as 5′UTR, CDS and even intron, promoter and enhancer, further both enhancing and suppressing the expression of target genes (Boudreau et al. 2014; Xiao et al. 2017). In this study, the binding sites were detected in CDS, 3′UTR and 5′UTR, and interestingly, 89% of binding sites are predominantly located within CDS (Figure S5). When comparing different species, miRNA binding sites vary dramatically and are species-specific, such as 3′UTR, 5′UTR, CDS and introns with 40% in CDS in human brain (Boudreau et al. 2014) and 70% in CDS in a green alga (Chung et al. 2017). The large proportion of CDS binding sites is likely due to the greater length of the coding sequence compared to UTR regions (Chung et al. 2017).

In addition, miRNAs regulate target genes in a 'multiple-to-multiple' manner. The effect of an individual miRNA on a target may be subtle, but targeting a set of genes within a specific pathway is a critical strategy to amplify the effects of miRNA (Ebert and Sharp 2012). Interestingly, numerous miRNAs have been shown to target specific functions in this study. For example, a total of 44 miRNAs were found to participate in ion channel regulation, while another 44 miRNAs were involved in regulating pathways related to FAA (Figure 7). In addition, one miRNA may regulate several strategies simultaneously; for example, cin-miR-4067-5p regulated both the FAA metabolism-related gene *CYP4F2.2* and the ion channel gene *CLCN6* (Figure 6 and Table S4).

Interestingly, both positive and negative regulation relationships have been observed when miRNAs bind to the 3′UTR, 5′UTR and CDS of target genes (Figure S6). Traditionally, 3′UTR binding was considered to suppress the expression of target genes. However, we observed instances of positive regulation through 3′UTR binding. For example, miR-4059-5p positively regulated the expression of *KCNK18* by targeting its 3′UTR (Figure S6E). In another study, miR-365 was shown to promote the expression of *HSP90AA1*, validated through in vivo and in vitro experiments, although the underlying mechanisms remained unclear (Chen et al. 2017). Furthermore, CDS binding typically leads to negative regulation on target genes. For instance, miR-4000b-5p negatively regulated *TRPC6* by binding to its CDS (Figure S6D). Previous studies have also observed that miR-932 targeted two sites in the CDS of *dnlg2*, resulting in the downregulation of *dnlg2* at both protein and RNA levels. The molecular mechanism may involve inducing mRNA degradation in combination with translation inhibition (Qian et al. 2016).

4.5 | Contributions of miRNA-Mediated Gene Regulations to Phenotypic Plasticity

Epigenetic processes, such as miRNA-mediated regulations of gene expression, are hypothesised to contribute to phenotypic plasticity in response to environmental changes in various ecological processes, including those involved in biological invasions (Mirouze and Paszkowski 2011). The high sensitivity of miRNAs to environmental perturbations and their crucial roles in fine-tuning gene expression make them prime candidates as dynamic regulators of phenotypic plasticity (Rubio-Somoza and Weigel 2011). According to our findings in this study, miRNA-mediated phenotypic plasticity may offer major advantages in rapidly responding to environmental changes. Firstly, miRNAs responded rapidly and reversibly to environmental stresses (Figure 2A–C), and such rapid response and subsequent regulation on gene expression can facilitate the survival of invaders under environmental changes. Secondly, miRNAs weave a complex regulatory network in response to environmental challenges (Pu et al. 2019). For example, in this study, nearly 25% of genes in the *C. robusta*'s genome were regulated by miRNAs in response to recurrent salinity stress (see Section 3.3). The diversified miRNA-mediated regulations on gene expression generate ample 'variation' in organisms, and as illustrated in this study, such variation encompasses dynamic miRNA expression patterns, diverse interplays with target genes, and ultimately functional adjustments to cope with salinity stresses (Figure 7 and Figures S3 and S4). This type of variation, devoid of genetic changes, empowers organisms to adeptly respond to environmental challenges, thus contributing to phenotypic plasticity such as enhanced survival capability. Moreover, such variation can buffer environmental challenges in specific populations that lack sufficient adaptive genetic variations, thereby aiding in their resilience and persistence. From a long-term evolutionary perspective, miRNA-generated variation in gene expression and regulatory networks serves as new raw materials for natural selection. This variation offers opportunities for organisms to explore and adapt to changing environments, ultimately driving evolutionary changes. This dual function underscores the significance of miRNAs in shaping the trajectory of short-term phenotypic plasticity and long-term evolution.

Indeed, miRNA-mediated phenotypic plasticity goes beyond immediate adaptive responses to environmental stresses, influencing broader ecological dynamics and long-term evolutionary trajectories (Levis and Pfennig 2016; Vinton et al. 2022). In the context of biological invasions, invasive species' ability to leverage miRNA regulatory networks for rapid and reversible responses enhances their capacity to establish and persist in novel habitats. This mechanism may confer a competitive advantage for invaders, especially in fluctuating or extreme environments where genetic adaptation alone may be insufficient (Xiang et al. 2023). Furthermore, miRNA-driven plasticity may facilitate niche expansion by enabling invaders to tolerate a broader range of environmental conditions, thereby potentially increasing invasion success.

From an evolutionary perspective, while miRNA-mediated gene regulation does not directly induce genetic mutations, it generates a reservoir of phenotypic variability upon which natural selection can act (i.e., the 'plasticity-first' hypothesis, Levis and Pfennig 2016). Over time, selective pressures from recurrent environmental changes may stabilise specific miRNA expression patterns, leading to the potential fixation of beneficial regulatory traits. However, the 'plasticity-first' hypothesis remains debated, primarily due to the lack of comprehensive tests that effectively identify the key components regulating phenotypic plasticity (Vinton et al. 2022). MiRNAs are an ideal target but remain the 'missing link' in testing the 'plasticity-first' hypothesis (Voskarides 2016). MiRNA-mediated plasticity could influence evolutionary rates by modulating gene expression in ways that promote survival under environmental stress, bridging the gap between short-term acclimatisation and long-term adaptation. These insights underscore the crucial role of miRNAs in shaping eco-evolutionary dynamics and suggest that their regulatory influence should be incorporated into predictive models across various ecological and evolutionary scenarios, such as species invasions, climate change responses and conservation management.

4.6 | Further Experimental Validation

In this study, we primarily employed in silico analysis to predict miRNA targets based on two sets of omics data using two widely recognised and effective methods. miRanda predicts targets through a three-step process: (i) sequence matching to identify maximal local complementarity between the miRNA and the putative target site, (ii) free energy calculation to estimate the stability of the potential RNA duplex and (iii) filtering of predicted targets based on evolutionary conservation (Enright et al. 2003). In contrast, RNAhybrid identifies energetically favourable binding sites, normalises them and applies estimated *p*-values to assess the statistical significance of predicted interactions (Rehmsmeier et al. 2004). These two methods employ different algorithms, complementing each other and mitigating individual shortcomings to reduce false positive rates.

Given the complexity of miRNA-mediated regulation, particularly where a single miRNA can target multiple genes and a single gene can be regulated by multiple miRNAs, experimentally validating such complex regulatory networks remains technically challenging, time-intensive and costly. Additionally, it requires specialised expertise, as well as the accumulation of appropriate experimental targets and materials such as competent cell lines from *Ciona*. Nevertheless, in silico analysis provides a crucial foundation for subsequent experimental validation (Beretta et al. 2016; Riolo et al. 2020). Indeed, the reliability of bioinformatics approaches is further supported by experimental validation across various taxa, including humans (Yang et al. 2015), fish (Chen et al. 2022), shellfish (He et al. 2024), plants (Yi et al. 2024) and insects (Mukherjee et al. 2021). However, further experimental validation of the findings in this study remains necessary. For instance, dual-luciferase reporter assays could be employed to confirm the predicted miRNA–target interactions, providing direct evidence of post-transcriptional regulation and strengthening the biological relevance of our in silico predictions.

5 | Conclusion

In this study, we employed recurrent osmotic stresses within a controlled system to replicate the environmental stress encountered by the model invasive ascidian *C. robusta*, and furthermore, we investigated the roles and mechanisms of miRNA-mediated gene regulations in facilitating rapid stress response. Recurrent osmotic stresses initially impacted both miRNA biogenesis and decay processes, with differential expressions observed in key functional genes such as *DGCR8*, *Drosha* and *XPO5* for miRNA biogenesis, and *TUT4* and *ZSWIM8* for miRNA decay. The increased miRNA abundance observed during stress stages suggests that the canonical biogenesis pathway for miRNAs, rather than degradation, is more significantly altered in response to recurrent salinity stresses. Through dynamic and diversified regulations on target genes, including diverse binding sites and positive/negative regulations, as well as the formation of stress memory, miRNAs effectively regulated three salinity coping strategies and associated genes, including water transport, ion transport and FAA metabolism. Key genes such as *CLIC5*, *SCN5A*, *KCNA4*, *NKA* and members of the *SLC* family were activated to influence ion concentrations, while the metabolism of eight FAAs was also affected. Notably, although *aquaporin* genes were directly regulated by miRNAs, no significant differential expression was observed. All these findings underscore the intricate, dynamic and efficient nature of miRNA-mediated gene regulations in response to environmental challenges.

Author Contributions

Weijie Yan: writing – review and editing, writing – original draft, methodology, investigation, formal analysis and data curation. **Ruiying Fu:** writing – review and editing, writing – original draft, methodology, investigation, formal analysis and data curation. **Xuena Huang:** writing – review and editing, writing – original draft, methodology, investigation, formal analysis, validation and funding acquisition. **Aibin Zhan:** writing – review and editing, writing – original draft, investigation, project administration and funding acquisition.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The miRNAomic and transcriptomic data have been deposited in the National Centre for Biotechnology Information (NCBI) Sequence Read Archive (SRA) database under the accession number PRJNA1120970 and PRJNA775866, respectively.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.